

RedKross Unified Threat Intelligence Platform

TRITON / TRISIS / HatMan

Malware Family Intelligence Report

ICS Safety System Targeted Malware | Nation-State Attribution | TLP:WHITE

Submitted By

Field	Detail
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Classification	TLP:WHITE — Unrestricted Distribution
Document Type	Malware Family Intelligence Page
Platform Module	Malware Family Module / ICS Threat Intelligence
Family Names	TRITON (primary) TRISIS HatMan
First Discovered	June 2017 (first plant shutdown); Publicly disclosed December 2017
Last Known Activity	2018 (US energy sector reconnaissance); Threat actor remains active as of 2022
Attribution	TEMP.Veles (FireEye) / XENOTIME (Dragos) — TsNIIKhM, Russian Federation
Primary Target	Safety Instrumented Systems (SIS) — Schneider Electric Triconex Tricon
Threat Level	CRITICAL — First malware publicly known to target SIS; potential for physical harm
Prepared For	RedKross Intelligence Platform — Malware Family Module

1. Executive Summary

TRITON (also tracked as TRISIS and HatMan) is a sophisticated, multi-stage malware framework designed to target Safety Instrumented Systems (SIS) in industrial control environments. It represents a critical milestone in ICS-targeted malware history: the first publicly known malware built explicitly to disable or subvert physical safety systems, creating conditions that could result in catastrophic industrial accidents, environmental damage, and loss of human life.

Discovered in mid-2017 following two safety-triggered plant shutdowns at a Middle East petrochemical facility (believed to be the Petro Rabigh refinery in Saudi Arabia, owned by Tasnee), TRITON was publicly disclosed in December 2017 by Mandiant and Dragos. It was purpose-built to interact with Schneider Electric Triconex Tricon safety controllers using a reverse-engineered implementation of the proprietary TriStation protocol. Attribution was subsequently made with high confidence to TsNIIKhM (Russian Central Scientific Research Institute of Chemistry and Mechanics) by both FireEye/Mandiant in 2018 and the U.S. Department of Justice in 2022.

Dragos and the U.S. Treasury Department's OFAC have characterised the TRITON actor as "the most dangerous threat activity publicly known," given its unique capability to neutralise the last line of physical safety defense in industrial environments.

2. Scope of This Report

- Full malware family history and discovery timeline
- Technical architecture, components, and infection chain
- Targeted systems, protocols, and vulnerable firmware versions
- Threat actor profiling (TEMP.Veles / XENOTIME / TsNIIKhM)
- Campaign history and known victim targeting
- MITRE ATT&CK; for ICS technique mappings

- IOC listing: hashes, filenames, network indicators
- YARA detection guidance and detection logic
- Mitigation and defensive countermeasures
- Historical timeline and new activity updates
- References and source attribution

3. Malware Family Overview

Field	Detail
Family Name	TRITON
Aliases	TRISIS, HatMan, Triton Framework
Malware Type	ICS Attack Framework / SIS Manipulation Tool
Platform	Windows (Engineering Workstation) + PowerPC (Triconex Controller)
Language	Python compiled via Py2Exe; PowerPC shellcode for controller payload stages
First Observed	June 2017 (plant safety trip); August 2017 (second shutdown / discovery)
Public Disclosure	December 14, 2017 — Mandiant / FireEye blog post
Attribution	TsNIIKhM — Russian Federation (High Confidence, FireEye 2018 / DOJ 2022)
Actor Aliases	TEMP.Veles (FireEye), XENOTIME (Dragos)
Primary Target	Schneider Electric Triconex Tricon MP3008 (firmware v10.0–10.4)
Secondary Target	Windows SIS Engineering Workstations
Known Victims	One confirmed: Middle East petrochemical (2017); US energy sector recon (2018)
Threat Impact	Physical destruction, industrial accident, environmental damage, loss of life
CVE / Zero-Day	Triconex firmware zero-day — privilege escalation for persistent firmware write

4. Discovery History & Timeline

Date	Event
~2014–2016	TEMP.Veles / TsNIIKhM believed to begin TRITON development and TriStation protocol reverse-engineering
June 2017	First TRITON deployment at Saudi petrochemical facility. Safety systems triggered an unplanned shutdown; initially attributed to a mechanical fault
August 2017	Second deployment at same facility. A bug in TRITON malware triggered another safety trip. Investigation reveals the malware and attacker presence
Dec 14, 2017	Public disclosure: Mandiant (FireEye) publishes TRITON report. Dragos publishes TRISIS analysis simultaneously. ICS-CERT releases HatMan advisory
2018 (Jan–Jul)	FireEye and Dragos publish detailed TTP profiles. XENOTIME (Dragos) observed conducting reconnaissance against US LNG and electric sector targets
May 2018	Reports emerge that TRISIS masterminds have expanded targeting to US industrial firms (CyberScoop / Chris Bing)
Oct 23, 2018	FireEye attributes TRITON development with high confidence to TsNIIKhM, a Russian government-owned research institution

Date	Event
Oct 23, 2020	US Treasury OFAC designates TsNIIKhM under CAATSA Section 224 for cybersecurity-undermining activities on behalf of the Russian government
Mar 24, 2022	US DOJ unseals indictments of three FSB officers and one TsNIIKhM employee (Evgeny Viktorovich Gladkikh) for TRITON deployment and US energy sector intrusion attempts
2022–present	FBI warns TsNIIKhM continues targeting global energy sector. CISA/NSA joint advisory issued. Threat actor remains active

5. Technical Architecture & Components

TRITON is a modular Python-based attack framework compiled to a Windows executable using Py2Exe. It consists of a dropper, a custom TriStation protocol implementation library, and two PowerPC shellcode payloads delivered to the Triconex controller. The following components have been publicly documented:

Component	Filename	Hash (MD5 unless noted)	Role
Main Framework	trilog.exe	dc81f383624955e0c0441734f9f1dabfe03f373c	Py2Exe-compiled dropper; masquerades as Triconex log analysis tool
Python Library	library.zip	1dd89871c4f8eca7a42642bf4c5ec2aa7688fd5c	All Python modules for TriStation comms and framework logic
Dropper Source	script_test.py	N/A (pre-compilation)	Original Python dropper before Py2Exe compilation
Stage-1 Payload	inject.bin	PowerPC shellcode	Masquerades as standard PowerPC program; injected into controller task table
Stage-2 Payload	imain.bin	PowerPC shellcode	Writes to firmware memory; hooks comms loop; enables persistent remote access
Actor Tool	cryptopp.dll	SHA-256: 32f5d0a4...5f0d51bbc MD5: 9a7234078559093e06c9d32148ed95a3	CryptoPP library used by TEMP.Veles actor toolset; placed in Office path

5.1 Infection Chain (Stage-by-Stage)

Stage 1 — Initial Access: TEMP.Veles gained initial access via spear-phishing or exploitation of internet-facing systems. Watering hole attacks and RDP-based lateral movement via jump-box infrastructure were used to traverse from IT to OT networks.

Stage 2 — IT/OT Pivot: Valid credentials and RDP jump boxes were used to move from the corporate IT network into the OT/ICS environment. The actor then gained access to the Windows-based SIS engineering workstation.

Stage 3 — Dropper Deployment: trilog.exe (the compiled TRITON framework) was copied to the SIS engineering workstation. The filename was chosen to masquerade as the legitimate Triconex log-analysis tool. The actor required the physical key on the Triconex controller to be in PROGRAM mode for firmware writes.

Stage 4 — TriStation Protocol Interaction: TRITON communicated with Triconex controllers over the TriStation UDP protocol (a proprietary, undocumented, authentication-less protocol that the actor reverse-engineered). TRITON queried controller status and used WRITE_TRICON_DATA to inject payload files.

Stage 5 — Payload Injection: inject.bin (Stage-1 shellcode) was injected into the Triconex program table, added to task execution so firmware runs it each cycle. inject.bin then loaded imain.bin (Stage-2), which exploited a Triconex firmware zero-day to write itself to firmware memory.

Stage 6 — Backdoor / Remote Control: imain.bin hooked the SIS communications main loop, providing persistent remote control of the safety controller. This could be used to force the SIS into an unsafe state (disabling emergency shutdowns) while a DCS-based attack created hazardous industrial conditions.

Stage 7 — Anti-Forensics / Cleanup: If TRITON detected an error, SafeAppendProgramMod attempted to reset the controller. If that failed, trilog.exe wrote a dummy program to controller memory — assessed as an anti-forensics measure to erase attacker code.

6. Threat Actor Profile

Field	Detail
FireEye / Mandiant Name	TEMP.Veles
Dragos Name	XENOTIME
MITRE ATT&CK; Group	G0088
Nation-State Attribution	Russian Federation
Sponsoring Entity	TsNIIKhM — Central Scientific Research Institute of Chemistry and Mechanics, Moscow
US Government Attribution	DOJ indictment (March 24, 2022); US Treasury OFAC designation (October 23, 2020)
Indicted Individual	Evgeny Viktorovich Gladkikh — TsNIIKhM employee charged with TRITON deployment and attempted US energy sector intrusions
Motivation	Nation-state espionage; pre-positioning for physical sabotage of critical energy infrastructure
Primary Sector Target	Oil & Gas, Petrochemical, Electric Utility
Geographic Targeting	Middle East (confirmed 2017); Europe, North America (observed 2018–2019 reconnaissance)
Other Tools	TEMP.Veles custom toolset; CryptoPP (cryptopp.dll); RDP lateral movement; credential harvesting tools
Initial Access Method	Watering hole attacks; spear-phishing; exploitation of internet-facing systems
Lateral Movement	RDP via jump boxes; valid account reuse across IT/OT boundary
Status	Active as of 2022 per FBI advisory; TsNIIKhM remains a sanctioned entity

7. Campaign Tracking

Campaign ID	Threat Actor	Period (UTC)	Country	Industry	Notes
CAMP-TRX-001	TEMP.Veles (XENOTIME)	Jun 2017–Aug 2017	Saudi Arabia	Petrochemical Oil & Gas	Confirmed SIS compromise at Petro Rabigh. Two safety trips. Malware discovered post-investigation.
CAMP-TRX-002	TEMP.Veles (XENOTIME)	Feb 2018–Jul 2018	United States	Oil & Gas LNG / Electric	Recon and intrusion attempts vs US-owned refinery infrastructure. No confirmed SIS compromise. Cited in 2022 DOJ indictment.
CAMP-TRX-003	XENOTIME	2019	Multi-region (EU/ME/NA)	Electric Utility ICS Vendors	XENOTIME expands to electric sector and ICS vendor/manufacturer targeting globally.

8. MITRE ATT&CK; for ICS Technique Mapping

Tech ID	Technique Name	Tactic	TRITON Usage
T0817	Drive-by Compromise	Initial Access	XENOTIME used watering hole websites targeting industrial employees
T0866	Exploitation of Remote Services	Lateral Movement	RDP jump boxes to traverse IT/OT boundary with valid credentials
T0883	Internet Accessible Device	Initial Access	Exploited internet-facing ICS/OT assets for network entry

Tech ID	Technique Name	Tactic	TRITON Usage
T0862	Supply Chain Compromise	Initial Access	Targeting of ICS vendors and manufacturers observed (2019 expansion)
T0843	Program Download	Execution	Downloaded malicious shellcode to Triconex controller via TriStation protocol
T0836	Modify Controller Tasking	Execution / Impair Process	inject.bin added to Triconex program task table; executed each firmware cycle
T0821	Modify Controller Software	Persistence / Inhibit Response	imain.bin wrote to firmware memory; exploited zero-day for privilege escalation
T0816	Device Restart/Shutdown	Inhibit Response Function	TRITON bugs caused unintended safety trips (two confirmed plant shutdowns)
T0851	Rootkit	Persistence / Defense Evasion	imain.bin hooked the firmware communications main loop for persistent access
T0872	Indicator Removal on Host	Defense Evasion	TRITON reset controller state or wrote dummy program on failure to erase attacker code
T0849	Masquerading	Defense Evasion	trilog.exe masqueraded as Triconex log tool; inject.bin as standard PowerPC program
T0855	Unauthorized Command Message	Impair Process Control	TriStation WRITE_TRICON_DATA commands sent without authentication to inject payloads
T0880	Loss of Safety	Impact	TRITON designed to reprogram SIS logic to allow unsafe process conditions to persist

9. IOC Knowledge Base

9.1 File Hashes

Hash Value	Type	File / Description	Source
dc81f383624955e0c0441734f9f1dabfe03f373c	MD5	trilog.exe — TRITON framework main dropper	Trellix / McAfee
1dd89871c4f8eca7a42642bf4c5ec2aa7688fd5c	MD5	library.zip — Python module bundle for TriStation comms	Trellix / McAfee
9a7234078559093e06c9d32148ed95a3	MD5	cryptopp.dll — CryptoPP tool; TEMP.Veles actor toolset	FireEye 2019
40361dad15b9b5ae2757a21d1ce6a61c3c37e891	SHA-1	cryptopp.dll	FireEye 2019
32f5d0a454c26e8aa6f4cad58f3782337cc97cfe2305bbfe564437e5f0d51bbc	SHA-256	cryptopp.dll	FireEye 2019 / Loki IOC

9.2 Key Filenames & Artifacts

Filename	Type	Notes
trilog.exe	Executable (PE/EXE)	TRITON framework dropper; masquerades as legitimate Triconex log analysis binary
script_test.py	Python Script	Original pre-compiled Python dropper source
library.zip	Python Archive	All Python modules for TriStation protocol communications
inject.bin	PowerPC Shellcode	Stage-1 payload; masquerades as standard Triconex PowerPC program
imain.bin	PowerPC Shellcode	Stage-2 payload; writes to firmware memory; hooks comms loop for persistence
cryptopp.dll	DLL	CryptoComputing library from TEMP.Veles toolset; planted in Office directory path to evade detection

9.3 Network / Protocol Indicators

Indicator Type	Value / Description	Notes
Protocol	TriStation UDP — undocumented Schneider Electric proprietary protocol	No authentication; TRITON uses custom reverse-engineered version
Attack Command	WRITE_TRICON_DATA (TriStation)	Core command used by TRITON to write payload to controller memory
Attack Command	GET_CONTROL_PROGRAM_STATUS (TriStation)	Used by TRITON to verify controller running state before injection
Attack Command	SafeAppendProgramMod (TriStation method)	Anti-forensics controller reset on injection failure
Infrastructure	RDP jump boxes (specific IPs not publicly disclosed)	XENOTIME IT/OT pivot infrastructure
Target Asset	Schneider Electric Triconex Tricon MP3008	Firmware versions 10.0 to 10.4 confirmed vulnerable

10. YARA & Detection Logic

The following YARA rules are based on publicly released detection logic from ICS-CERT, Dragos, and community research. Deploy on Windows engineering workstations in ICS environments.


```

rule TRITON_Framework_Dropper {
  meta:
    description = "Detects TRITON ICS attack framework (trilog.exe)"
    family      = "TRITON / TRISIS / HatMan"
    tlp         = "WHITE"
  strings:
    $py2exe1 = "script_test"      ascii
    $ts1      = "TriStation"       ascii nocase
    $ts_cmd1  = "WRITE_TRICON_DATA"  ascii
    $ts_cmd2  = "GET_CONTROL_PROGRAM"  ascii
    $ts_cmd3  = "SafeAppendProgramMod"  ascii
    $payload1 = "inject.bin"        ascii
    $payload2 = "imain.bin"         ascii
  condition:
    uint16(0) == 0x5A4D and filesize < 10MB and
    (2 of ($ts_cmd*) or all of ($payload*) or
    ($py2exe1 and $ts1))
}

rule TRITON_TriStation_Comms_Library {
  meta:
    description = "Detects TRITON Python TriStation communications library"
  strings:
    $ts1 = "TriStation"      ascii
    $ts2 = "triconex"        ascii nocase
    $c1  = "WRITE_TRICON_DATA"  ascii
    $c2  = "READ_TRICON_DATA"  ascii
    $c3  = "AppendProgramMod"  ascii
  condition:
    2 of ($ts*) and 1 of ($c*)
}

```

Detection Notes: Tenable provides detection via plugin ID 59275 (Malicious Process Detection). Nozomi Networks provides a Wireshark dissector for the TriStation protocol for network-level detection. Schneider Electric issued a firmware patch for the Triconex zero-day exploited by imain.bin.

11. Victimology & Target Profile

Attribute	Detail
Confirmed Victim (2017)	Middle East petrochemical facility — believed to be Petro Rabigh (Saudi Arabia, Tasnee / Saudi Aramco JV)
Victim Sector	Petrochemical / Oil & Gas (confirmed); Electric Utility (reconnaissance)
Victim Countries	Saudi Arabia (confirmed, 2017); United States (attempted, 2018); Europe (observed reconnaissance)
Target Asset Class	Safety Instrumented Systems (SIS) — specifically Schneider Electric Triconex Tricon controllers
Target Process	Sulfur recovery units and burn management systems (per academic analysis)
Expanded Targeting (2018–19)	US LNG terminals, electric utilities, ICS vendors and OT manufacturers across Middle East, Europe, North America
Why SIS?	SIS is the last automated safety barrier in industrial facilities. Defeating it removes all automated emergency shutdown capability

12. Infrastructure Notes

- TEMP.Veles used RDP jump-box infrastructure to bridge IT and OT segments, leveraging valid (stolen) credentials throughout.
- The TriStation protocol operates over UDP; no public IPs or C2 domains have been attributed to TRITON's network infrastructure in open-source reporting.
- The actor used watering hole websites targeting industrial sector employees as an initial access vector.
- cryptopp.dll was placed at a Microsoft Office directory path (e.g. Office12\) to evade detection among legitimate Office files.
- No external C2 for the SIS controller backdoor has been publicly confirmed; imain.bin communicated via the hooked TriStation loop rather than outbound internet connections.
- The Python-based framework was compiled to an EXE (Py2Exe) to reduce Python dependency artifacts and blend with legitimate tools.
- Schneider Electric confirmed the zero-day vulnerability and issued remediation patches. The TriStation protocol's lack of authentication remains a design risk for legacy deployments.

13. Behavioral Characteristics & Unique Artifacts

13.1 Unique File System Artifacts

- trilog.exe in the SIS engineering workstation directory (non-standard path for this legitimate tool name)
- library.zip alongside trilog.exe — Python module archive not normally present on engineering workstations
- inject.bin and imain.bin as standalone binary files on the engineering workstation
- cryptopp.dll in Microsoft Office directory paths (anomalous DLL placement)
- script_test.py if pre-compilation artifacts are recovered

13.2 Unique Behavioral Indicators

- TriStation protocol traffic originating from unexpected hosts (not the authorized engineering workstation)
- TriStation WRITE_TRICON_DATA commands followed by periodic GET_CONTROL_PROGRAM_STATUS polling in short intervals — indicative of payload injection loop
- Unplanned SIS controller safety trips with no corresponding process anomaly (as observed June and August 2017)
- RDP sessions crossing the IT/OT boundary to engineering workstations from non-standard jump boxes
- Triconex controller in PROGRAM mode outside of scheduled maintenance windows
- Py2Exe artifacts (.pyd files, python27.dll) present on SIS engineering workstations

14. Persistence, Privilege Escalation & Exfiltration

Category	Detail
Persistence — Workstation	trilog.exe installed on SIS engineering workstation; re-runs and re-injects on demand
Persistence — Controller	imain.bin writes to Triconex firmware memory (not user area), surviving user-area re-downloads; hooks firmware communications main loop
Privilege Escalation	Triconex firmware zero-day exploited by imain.bin to escalate from user-area to firmware-level write access
Defense Evasion	Filename masquerading; dummy program write on failure; valid credentials throughout; Py2Exe compilation to minimise Python artifacts
Exfiltration	Not primary objective; TRITON is a pre-positioning/sabotage tool. No confirmed exfiltration of process data
Potential Impact	Disabling all automated emergency shutdown capability; enabling physical destruction when combined with DCS-based process manipulation

15. Mitigation & Defensive Countermeasures

The following mitigations are drawn from CISA Advisory AA22-083A, ICS-CERT guidance, Schneider Electric recommendations, and community best practices:

- **Apply Schneider Electric Triconex patch:** Install vendor-released firmware patch addressing the zero-day privilege escalation exploited by TRITON's imain.bin payload.
- **Physical key management:** Ensure Triconex controllers remain in RUN (not PROGRAM) mode except during authorized maintenance. Control physical access to the key.
- **Network segmentation:** Enforce strict IT/OT network separation. Eliminate unnecessary RDP paths into engineering workstations. Block unauthorized TriStation traffic via firewall rules.
- **Engineering workstation hardening:** Restrict which hosts may communicate with SIS controllers via TriStation. Monitor for unexpected TriStation traffic from non-authorised workstations.
- **Application allowlisting:** Prevent unauthorized executables (trilog.exe look-alikes, Py2Exe artifacts) from running on SIS engineering stations.
- **YARA scanning:** Deploy YARA rules (Section 10) on engineering workstations. Use Tenable plugin ID 59275 for Nessus-based detection.
- **TriStation traffic monitoring:** Deploy the Nozomi Networks Wireshark dissector or equivalent ICS network monitoring to detect anomalous TriStation commands outside maintenance windows.
- **Credential hygiene:** Audit and rotate OT/ICS account credentials. Enforce MFA where feasible. Monitor for credential reuse from IT to OT environments.
- **Jump box monitoring:** Log and alert on RDP sessions to engineering workstations. Review hop patterns crossing the IT/OT boundary.
- **SIS audit logging:** Enable and monitor Triconex audit logs for unexpected program downloads, mode changes, or status polling sequences.
- **Vendor guidance:** Review CISA ICS Advisory for Schneider Electric Triconex Tricon (Update B) and CISA MAR HatMan (Update B).

16. Related Malware & ICS Threat Comparisons

Malware	Year	Target System	Actor	Comparable Aspects
Stuxnet	2009–2010	Siemens S7 PLCs (uranium centrifuges)	USA / Israel (attributed)	First major ICS malware; PLC firmware manipulation; physical damage intent
Industroyer / CrashOverride	2016	IEC 104/101/61850 RTUs; circuit breakers	Sandworm (Russia)	ICS protocol abuse; grid disruption; Ukraine power outages 2016
TRITON / TRISIS	2017	Schneider Electric Triconex SIS controllers	TEMP.Veles / TsNIICKhM (Russia)	First SIS-targeting malware; firmware backdoor; potential for physical harm
Industroyer2	2022	IEC 104 RTUs — Ukrainian grid	Sandworm (Russia)	Evolution of Industroyer; deployed during Ukraine conflict
PIPEDREAM / INCONTROLLER	2022	Schneider Electric, Omron OT assets	Chernovite (Dragos)	Next-generation ICS framework; multi-vendor OT support; overlaps TRITON's SIS scope

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